Sangeon Jeon

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Education

Georgia Institute of Technology | Atlanta, GA
Master of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 *January 2025 – December 2025(Expected)*
Bachelor of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 *January 2023 – December 2024*

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
Bachelor of Science in Electrical Engineering, GPA 3.95 /4.3 *February 2018 – February 2025*

Work Experience

Dong Il Vision | Anyang, South Korea *June – August 2023*
Intern / Software Engineer

- Developed a C# WPF labeling tool to annotate images for machine learning.
- Contributed to building a computer vision system that differentiated products from competitors.

Research

Xiaoli Ma Lab @ Gatech | Atlanta, GA (Advisor: Prof. Xiaoli Ma) *Summer 2025 - Present*

- Conducting MATLAB simulations on Orthogonal Chirp Division Multiplexing (OCDM), a next-generation transmission scheme.

SIC-X Lab @ KAIST | Daejeon, South Korea (Advisor: Prof. Jinseok Choi)
Joint estimation of channel and mismatch for OFDM receiver with TIADCs *Fall 2024 - Present*

- Designed algorithms to estimate gain, timing, and offset mismatches in Time-Interleaved ADCs (TIADCs) under OFDM.
- Built a suboptimal maximum likelihood estimator with polynomial rooting, validated via MATLAB simulations.
- Achieved 84% reduction in computation time compared to existing methods while improving accuracy.
- Enhanced receiver calibration for high-speed OFDM systems, reducing mismatch-induced distortion in wideband communication channels.
- Currently preparing a research paper for publication.

Project

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
KAIST Maker-a-thon (Topic: Item for the Elderly Living Alone) *Summer 2024*

- Participated in three weeks of development, brainstorming and prototyping an item suitable for the topic using materials and equipment provided by KAIST makerspace. Led a team of four as the leader and won the top award with $2000.
- Proposed an RFID token system to manage the expiration dates of food in refrigerators. Innovated the pre-existing RFID refrigerator system by installing an RFID reader on a token container instead of the refrigerator.

Georgia Institute of Technology | Atlanta, GA
Senior Design Project: Plant Physiology *Spring 2024*

- Collaborated with Vivent (Switzerland) to build a system for recording and replaying the action potential of a Venus flytrap, achieving successful trap closure using only electrical stimulation.
- Designed the overall project blueprint, including software specifications, signal acquisition, and playback methodology.
- Developed a diagnostic framework based on impulse response analysis to assess plant state, reconstruct native signals, and generate artificial action potentials.
- Implemented a Seeeduino–MATLAB communication protocol with a custom GUI, and optimized algorithms to correct time delays, enabling reliable signal acquisition up to 100 Hz.

Awards

Roger Webb Award ECE Senior Scholar Awards (2025), 1st Prize at KAIST Maker-a-thon (2024), Dean's List (5 semesters)

Relevant Coursework

KAIST: Signal and System, Probability and Introductory Random Processes, Linear Algebra, Communication Engineering
Gatech: Fundamental of DSP, Application of DSP, Random Processes, Adaptive Filtering

Skills

Programming: MATLAB, Python (Pytorch, PyMC), C++
Hardware: Oscilloscopes, Function Generators, Multimeters, ATmega128A, Xiao Seeeduino
Languages: Korean (native), English (conversational), Japanese (intermediate), Chinese(beginner)